Modified Problem Statement –

a Doubly Linked List (DLL) variation with structural refinements, and a Circular Doubly Linked List (CDLL) variant, which fits graphics loops by letting you cycle from top to bottom seamlessly.

Objectives –

- **Manage Stacking Contexts Flexibly:** Implement structural representations of graphic project layers using two alternative approaches: a linear sequence with definite boundaries via a Doubly Linked List (DLL), and a continuous loop environment via a Circular Doubly Linked List (CDLL).
- **Optimize Top & Bottom Boundary Access:** Ensure instant insertion operations at the absolute background layer depth (O(1) at head) and foreground layout depth (O(1) at tail or head->prev) without sweeping the entire data frame layout.
- **Maintain Dynamic Rendering Order:** Build standard algorithms to swap node structural pointers, shifting layers forward safely without risking memory isolation bugs or breaking boundary constraints.
- **Automate Garbage Deallocation:** Identify, detach, and permanently free memory layouts belonging to invisible or muted layers (isActive == false) while keeping tracking integrity stable.